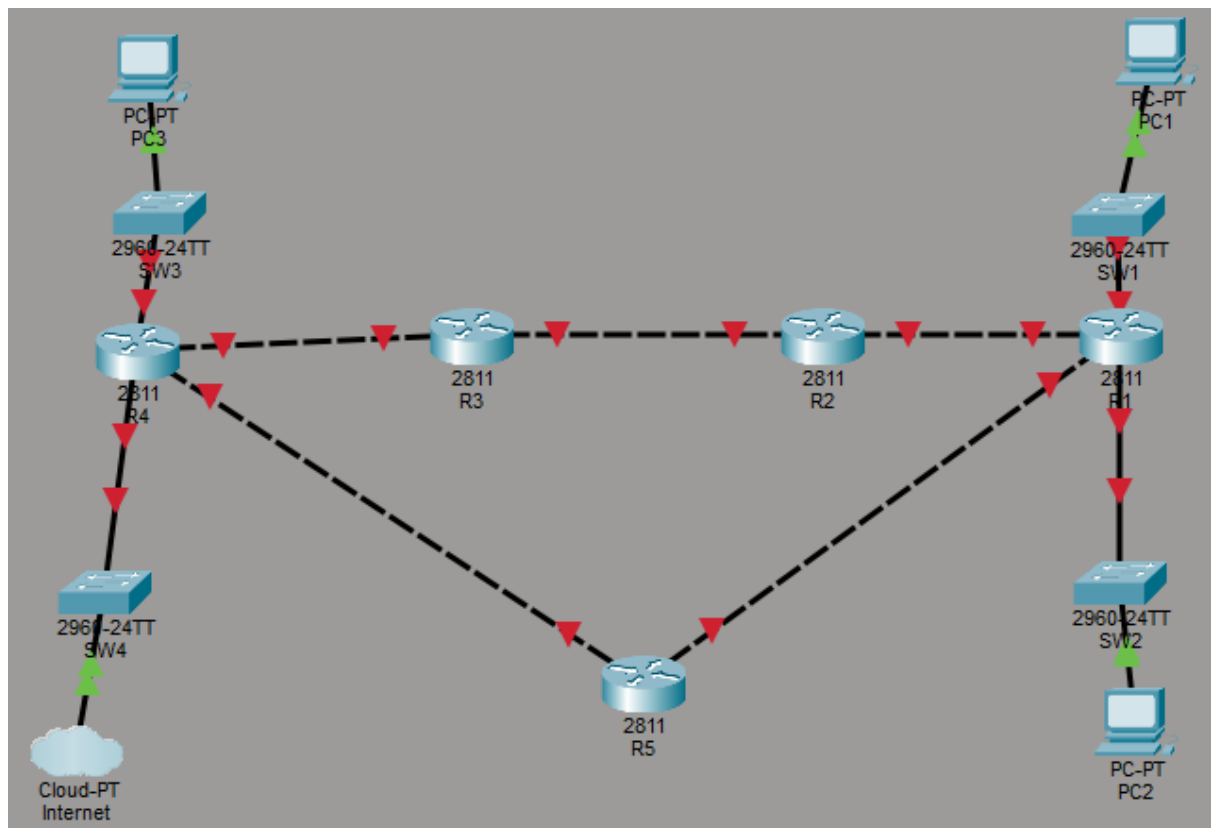


Week 7 cisco packet tracer:

1.) Topology:



2.) Configuring hostnames:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#end
R1#
```

3.) Configuring R1:

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fa0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface fa0/1
R1(config-if)#ip address 10.0.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface fa1/0
R1(config-if)#ip address 10.0.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface fa1/1
R1(config-if)#ip address 10.0.3.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#end
--
```

4.) R1 ip interface brief:

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.1        YES manual up          up
FastEthernet0/1          10.0.1.1        YES manual up          up
FastEthernet1/0          10.0.2.1        YES manual up          up
FastEthernet1/1          10.0.3.1        YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
```

5.) Ip route:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1
```

6.) Pinging PC1 TO PC2:

```
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7.) Pinging PC1 TO PC3:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Request timed out.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

7.) Configuring Routers:

```

R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface fa0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface fa0/1
R2(config-if)#ip address 10.1.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#end

```

```

R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface fa0/0
R3(config-if)#ip address 10.1.1.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface fa0/1
R3(config-if)#ip address 10.1.0.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#end

```

```

R4>enable
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#interface fa0/0
R4(config-if)#ip address 10.1.1.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface fa0/1
R4(config-if)#ip address 10.1.2.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface fa1/0
R4(config-if)#ip address 10.1.3.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#end

```

8.) Verifying PC3 gateway:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

9.)

Configuring routes:

```

R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.2
%Invalid next hop address (it's this router)
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.3.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.3.0 255.255.255.0 10.1.0.1
R2(config)#end

```

10.)Verifying connectivity after configuring routes:

```

C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>

```

```

C:\>tracert 10.1.2.10

Tracing route to 10.1.2.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  0 ms    *       0 ms    10.0.1.1
  2  *       0 ms    *       Request timed out.
  3  0 ms    *       0 ms    10.0.1.1
  4  *       0 ms    *       Request timed out.
  5  0 ms    *       0 ms    10.0.1.1
  6  *       0 ms    *       Request timed out.
  7  0 ms    *       0 ms    10.0.1.1
  8  *       0 ms    *       Request timed out.
  9  0 ms    *       0 ms    10.0.1.1
 10  *       0 ms    *       Request timed out.
 11  0 ms    1 ms    *       Request timed out.
 12  0 ms    *       0 ms    10.0.1.1
 13  *       0 ms    *       Request timed out.
 14  0 ms    *       0 ms    10.0.1.1
 15  *       0 ms    *       Request timed out.
 16  0 ms    *       3 ms    10.0.1.1
 17  *       0 ms    *       Request timed out.
 18  0 ms    *       0 ms    10.0.1.1
 19  *       0 ms    *       Request timed out.
 20  0 ms    *       0 ms    10.0.1.1
 21  *       0 ms    *       Request timed out.
 22  0 ms    *       0 ms    10.0.1.1
 23  *       0 ms    *       Request timed out.
 24  0 ms    *       0 ms    10.0.1.1
 25  *       0 ms    *       Request timed out.
 26  0 ms    *       0 ms    10.0.1.1
 27  *       0 ms    *       Request timed out.
 28  0 ms    *       0 ms    10.0.1.1
 29  *       0 ms    *       Request timed out.
 30  0 ms    *       0 ms    10.0.1.1

```

11.)Removing R1 static routes:

```

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#no ip route 10.1.0.0 255.255.255.0 10.0.0.2
%No matching route to delete
R1(config)#no ip route 10.1.1.0 255.255.255.0 10.0.0.2
%No matching route to delete
R1(config)#no ip route 10.1.2.0 255.255.255.0 10.0.0.2
%No matching route to delete
R1(config)#no ip route 10.1.3.0 255.255.255.0 10.0.0.2
%No matching route to delete
R1(config)#
R1(config)#end

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1

```

12.)task 62:

```

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.1.0.0 255.255.0.0 10.0.0.2
R1(config)#end

```

12.)Verifying after summary route:

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1
S       10.1.0.0/16 [1/0] via 10.0.0.2

R1#ping 10.1.2.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.10, timeout is 2 seconds:
...!!!
Success rate is 40 percent (2/5), round-trip min/avg/max = 1/1/1 ms

R1#ping 10.1.2.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/12 ms

```

13.)Task 70:

```

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1
S       10.1.0.0/16 [1/0] via 10.0.0.2
S       10.1.3.0/24 [1/0] via 10.0.3.2
S*      0.0.0.0/0 [1/0] via 10.0.0.2

```

Task 71: yes R1 automatically has routes for every network that is connected to the interface.

Task 72: because R1 no longer has a route to PC3 network>

Task 73: because the networks share the same first 16 bits.